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Infineon Technologies

Hiring Team

Application: Master Thesis, AI Based Condition Monitoring for Drive Systems

Dear Hiring Team,

I am applying for the Master Thesis on AI based condition monitoring for drive systems at Infineon Technologies. The brief covering synthetic data generation, machine learning model development and training, and validation under realistic operating conditions, with a specific use case around DC Link capacitor degradation, aligns closely with the work I have been shipping.

At eRay GmbH I built a recursive time series pipeline forecasting chlorophyll a, turbidity, pH, and dissolved oxygen for a German lake, benchmarking six models head to head and using CatBoost multi quantile regression for asymmetric 80 percent prediction intervals. I enforced strict anti leakage rules, reconstructed missing winter readings with MICE imputation, and wrapped the pipeline in an orchestrator with gate checks. That is exactly the discipline required for reliable condition monitoring signals off drive systems.

In CreditIQ I designed and validated a machine learning pipeline against strict evaluation criteria, cutting the false negative rate from 44 percent to 16.7 percent while holding accuracy at 75 percent and backing everything with unit tests at 100 percent branch coverage. In my Real Time Flight Tracking Data Pipeline I processed more than 128 thousand records with PySpark on Google Cloud and orchestrated the whole system with Apache Airflow, refreshing every 15 minutes. That is a solid foundation for the data preprocessing, feature extraction, and labelling side of the thesis.

I am proficient in Python and scikit learn, comfortable in MATLAB and Simulink at the level required, and hold the NVIDIA Building LLM Applications With Prompt Engineering certificate. I would be glad to shape the thesis with my professor at SRH Heidelberg and align on the drive system context with the Application and AI teams.

Kind regards,

Rahul Rawat